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Regional seawater-chemistry controls on ocean alkalinity enhancement efficiency: Carbonate precipitation risk under pH-equilibrated OAE

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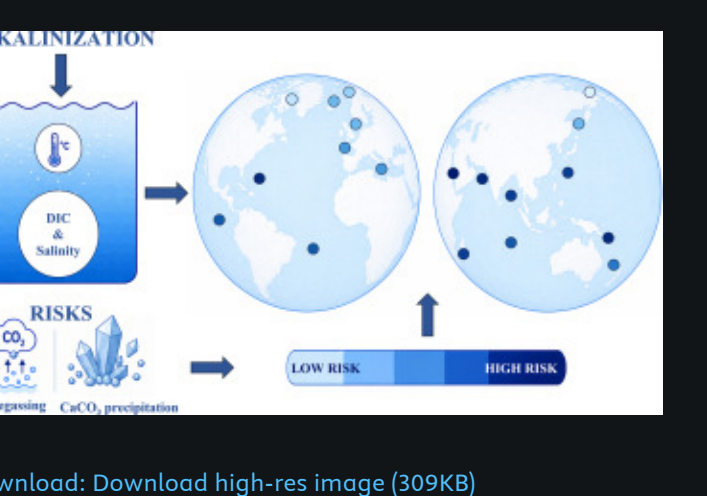
Highlights

- Simulations of seawater conditions in different sites, after pH-equilibrated ocean alkalinity enhancement (OAE).
- Regional seawater chemistry controls aragonite supersaturation (Ω_{Ar}) and long-term stability of pH-equilibrated OAE.
- Excessive dissolved inorganic carbon increases precipitation risk and CO_2 degassing, mainly in warm, saline regions.
- Ω_{Ar} screening supports safer site selection and monitoring for OAE.

Abstract

Ocean alkalinity enhancement (OAE) is a marine carbon dioxide removal (mCDR) strategy that increases seawater alkalinity and promotes atmospheric CO_2 uptake. However, the stability of stored carbon remains uncertain, particularly when carbonate saturation favors secondary calcium carbonate precipitation. In this study, we evaluate how regional seawater chemistry controls aragonite saturation state (Ω_{Ar}) under pH-equilibrated OAE conditions, in which seawater is enriched in dissolved inorganic carbon (DIC) while pH remains close to the natural seawater value. Near-surface seawater conditions were selected from ocean regions using GLODAPv2.2023 data, and Ω_{Ar} was calculated for baseline conditions and scenarios of +500, +1000, and +2000 μmol kg^{-1} added DIC. The data show that the carbonate chemistry is not uniform among regions. Thus, although the same DIC additions were adopted in all regions, the post-treatment Ω_{Ar} differed because of the non-uniform initial seawater chemistry. High temperature, high salinity, and high pH regions feature higher aragonite supersaturation and move toward the $\Omega_{Ar} = 7$ threshold more easily, whereas colder or lower salinity regions remain far from this level. After a DIC addition of 1000 μmol kg^{-1}, most regional median Ω_{Ar} values remain below $\Omega_{Ar} = 7$. In contrast, after a DIC addition of 2000 μmol kg^{-1}, several sites fall into dangerously higher supersaturation conditions. These findings indicate that carbonate supersaturation and secondary precipitation risk depend strongly on the receiving seawater chemistry. Therefore, OAE should not be evaluated using a universal alkalinity dosage. An assessment of the region-specific carbonate chemistry is necessary before selecting the treatments and deployment sites.

Graphical abstract



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Keywords

Ocean alkalinity enhancement; Precipitation risk; Long-term stability; OAE global performance; Carbonate minerals

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1. Introduction

Ocean alkalinity enhancement (OAE) is a marine carbon dioxide removal (mCDR) strategy based on accelerating carbonate weathering-driven alkalinity that naturally helps mitigate atmospheric CO_2 over long timescales (Rau and Caldeira, 1999; Caldeira and Rau, 2000; Renforth and Henderson, 2017; Kheshgi, 1995). In OAE, alkalinity can be added to seawater either directly or after a CO_2-equilibration via a pretreatment step (Kheshgi, 1995; Renforth and Kruger, 2013; Rau and Caldeira, 1999; Caldeira and Rau, 2000; Caserini et al., 2021; Kirchner et al., 2020; Chou et al., 2015; Eisaman et al., 2023; Varliero et al., 2024). The added alkalinity shifts the marine carbonate system toward bicarbonate (HCO_3^-) and carbonate (CO_3^{2-}) ions, lowers surface ocean pCO_2, and promotes atmospheric CO_2 uptake, while, as a secondary co-benefit, the alkalinity increase can also counteract ocean acidification (Renforth and Henderson, 2017; Zhou et al., 2025).

The performance and safety of OAE, however is not controlled only by the amount of added alkalinity. At the carbonate-system level, the theoretical CO_2 uptake per unit added alkalinity depends on hydrographic and carbonate-chemistry conditions, including temperature, salinity, pCO_2, and the initial state of the marine carbonate system (Renforth and Henderson, 2017; Burt et al., 2021; Humphreys et al., 2018). After complete air-sea CO_2 re-equilibration, the local maximum OAE efficiency has been estimated to vary between about 70% and 90% over the global ocean, depending mainly on temperature and background carbonate chemistry (Renforth and Henderson, 2017; Burt et al., 2021; Humphreys et al., 2018). The local efficiency of OAE can also be affected by circulation, ventilation, mixed-layer dynamics, seasonality, and air-sea gas exchange, which control how long added alkalinity remains in surface waters and where the induced CO_2 uptake occurs (Zhou et al., 2025; Burt et al., 2021; He and Tyka, 2023).

Most previous global and near-coastal modelling studies have evaluated OAE efficiency mainly in terms of atmospheric CO_2 uptake after alkalinity addition (Zhou et al., 2025; Burt et al., 2021; He and Tyka, 2023). In this framework, computed efficiency is strongly influenced by whether added alkalinity remains in surface waters long enough to equilibrate with atmospheric CO_2 (Zhou et al., 2025; Burt et al., 2021; He and Tyka, 2023). Incomplete air-sea CO_2 equilibration can reduce carbon removal efficiency when excess alkalinity is transported away from the surface before full equilibration occurs. Zhou et al. (Zhou et al., 2025) described this response using two characteristic timescales: an initial, relatively rapid surface equilibration phase, followed by a slower phase linked to the re-emergence of alkalinity that had previously been subducted.

These uptake-based efficiency estimates, however, do not fully address whether the treated seawater remains chemically stable after alkalinity enhancement. Alkalinity addition increases carbonate ion concentration and raises the saturation state of seawater with respect to carbonate minerals. Similarly, in the case of pH-equilibrated OAE, DIC enrichment at fixed pH can increase the aragonite saturation state (Ω_{Ar}). If Ω_{Ar} or local supersaturation becomes sufficiently high, secondary calcium carbonate precipitation may be triggered, transferring alkalinity and inorganic carbon from solution into solid carbonate phases and reducing the durability and efficiency of the intended carbon storage (Moras et al., 2022; Hartmann et al., 2023; Ringham et al., 2024; Suitner et al., 2024; Varliero et al., 2024; Jamali Alamooti et al., 2026). Therefore, carbonate-system stability and precipitation risk represent a complementary aspect of OAE performance that is not captured by long-term atmospheric CO_2 uptake efficiency alone.

In this study, we focus on the pH-equilibrated OAE method, in which seawater is enriched in DIC while pH is maintained close to the natural seawater value. This formulation should not be interpreted as DIC-only addition, which would lower pH. Rather, it represents a DIC-enriched carbonate-system condition at the constant pH, requiring a corresponding increase in TA relative to untreated seawater. This carbon-enriched solution represents a pretreatment pathway before discharge, such as CO_2 equilibration of alkalinized seawater, as in the previously tested pH-equilibrated process where CO_2 is first added to seawater and $Ca(OH)_2$ slurry is then introduced into the CO_2-enriched seawater to produce a bicarbonate-rich solution at seawater pH (Varliero et al., 2024; Jamali Alamooti et al., 2026), or a state reached after alkalinity addition and subsequent air-sea CO_2 uptake. Our previous mesocosm study in the Gulf of La Spezia showed that local salinity, temperature, pH, and DIC variables can strongly influence Ω_{Ar} and therefore the long-term stability of pH-equilibrated OAE (Jamali Alamooti et al., 2026). Building on this knowledge, the objective of this work is to evaluate how regional seawater chemistry controls the increase in Ω_{Ar} under pH-equilibrated DIC enrichment and identify ocean settings where this seawater treatment can be applied in safer conditions and more efficiently.

2. Methodology

To evaluate the effect of regional seawater variability on pH-equilibrated OAE, this study used carbonate-chemistry and hydrographic data from the Global Ocean Data Analysis Project, version 2.2023 (GLODAPv2.2023) (Lauvset et al., 2024). Near-surface observations from selected seas and regions within the Arctic, Atlantic, Indian, and Pacific Oceans were used as input parameters for carbonate system calculations. The analysis focused on Ω_{Ar}, which provides an indicator of how close seawater is to carbonate mineral supersaturation and therefore addresses the potential risk of secondary carbonate precipitation after DIC addition. Ω_{Ar} is defined as (Zeebe and Wolf-Gladrow, 2001):

$$\Omega_{Ar} = \frac{[Ca^{2+}]\,[CO_3^{2-}]}{K_{SP}} \qquad (1)$$

where K_{sp} is the solubility product of aragonite.

Higher Ω_{Ar} values indicate stronger supersaturation with respect to aragonite. Here, Ω_{Ar} was used as a carbonate system screening parameter for comparing baseline seawater conditions and pH-equilibrated OAE scenarios across contrasting oceanographic settings. Moras et al. (Moras et al., 2022) reported no $CaCO_3$ precipitation at Ω_{Ar} around 5, while aragonite precipitation occurred above Ω_{Ar} around 7. Based on this experimental reference, $\Omega_{Ar} = 7$ was used here only as a visual indicator to distinguish lower- and higher-supersaturation zones, without implying that this is the precipitation threshold. This study therefore uses Ω_{Ar} as a screening parameter to show the shift in supersaturation after treatment, not as a kinetic precipitation model. The actual onset of $CaCO_3$ precipitation depends on additional factors, including nucleation conditions, organic matter, phosphate, Mg^{2+} effects, suspended particles, and hydrodynamic mixing and dilution.

GLODAPv2.2023 is a merged and bias-adjusted product of ocean biogeochemical bottle data from hydrographic cruises, with a focus on seawater inorganic carbon chemistry and related variables. The database includes more than 1.4 million seawater samples from 1108 cruises. According to Lauvset et al. (Lauvset et al., 2024), the bias-adjusted data are internally consistent to better than 0.005 in salinity, 4 μmol kg^{-1} for dissolved inorganic carbon (DIC) and total alkalinity (TA), and 0.01–0.02 for pH, depending on region. For the present analysis, sampling date, latitude, longitude, depth, temperature, salinity, DIC, TA, and pH were extracted. The pH variable used here was pH on the total scale at in situ temperature and pressure. These variables were selected because they define the carbonate-system state required to calculate Ω_{Ar} under the corresponding hydrographic conditions.

The data files were first screened to retain only the variables required for the carbonate-system calculations and regional selection. Missing or invalid numerical values, including values coded as −999 or −9999, were converted to missing values. Observations with missing values in the key carbonate-chemistry or hydrographic variables were then removed. The retained variables included sampling date, latitude, longitude, depth, temperature, salinity, DIC, TA, and pH. Because pH-equilibrated OAE would mostly be implemented in surface or near-surface waters, and because air-sea exchange occurs at the ocean surface, only near-surface observations were retained for the analysis. A depth threshold of ≤10 m was applied to define the near-surface dataset, consistent with the surface layer directly involved in air-sea exchange in OAE modelling studies (Zhou et al., 2025).

Selected seas and ocean regions were then defined within the Arctic, Atlantic, Indian, and Pacific Oceans using latitude-longitude boundaries. These regions were chosen to cover a wide range of natural seawater conditions, including differences in temperature, salinity, pH, and DIC. The geographic boxes and the selected observations are shown in Fig. 1. For each sea or region, 15 representative near-surface data points were selected based on the pH distribution of the filtered data because pH was assumed to remain fixed before and after pH-equilibrated OAE. Specifically, five observations closest to the minimum pH, five observations closest to the mean pH, and five observations closest to the maximum pH were retained. This selection was used to represent low-pH, average-like-pH, and high-pH conditions within each region. The selected 15 points were intended to span the observed carbonate-chemistry range of each region, not to provide a seasonally weighted climatological mean. Although the selection was based on pH, the co-observed temperature, salinity, DIC, TA, depth, and sampling date of each selected GLODAPv2.2023 observation were retained in the calculations. Therefore, hydrographic variability associated with seasonal conditions, river influence, or upwelling was partly preserved in the selected carbonate-system states. Repeated or overlapping geographic positions were retained when they corresponded to distinct observations, depths, dates, or carbonate chemistry conditions. No additional filter was applied based on sampling year or season. After this screening and the near-surface depth filter, no additional salinity or hydrography outlier filter was applied before selecting the minimum and maximum pH points. Possible river-influenced, upwelling-related, or other localized near-surface signals were retained when they had complete carbonate-system information, because the aim was to cover the observed carbonate-system within each region.

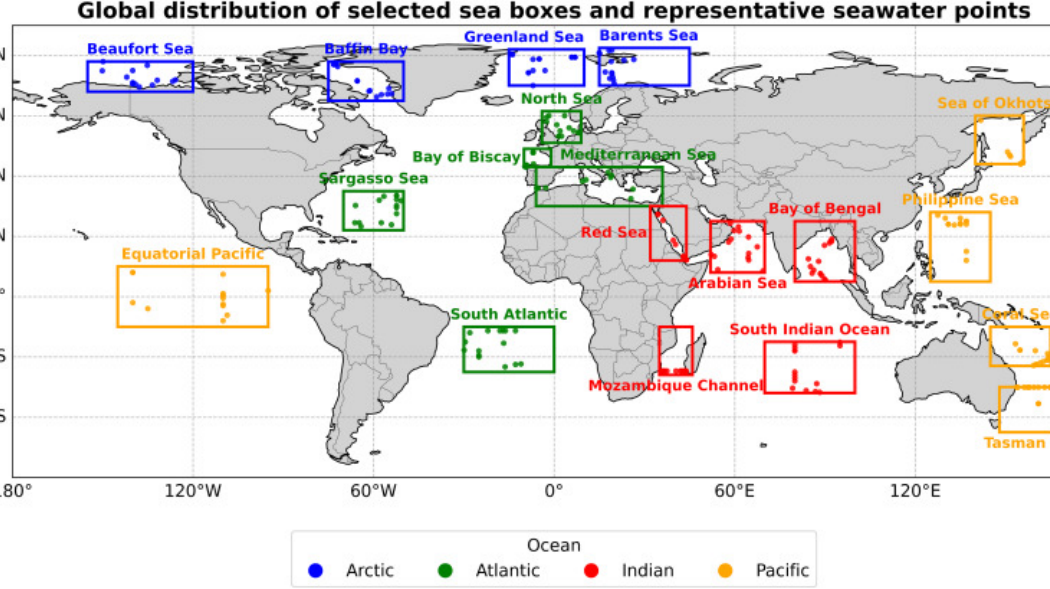

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Fig. 1. Geographic distribution of selected sea and regional boxes and representative near-surface GLODAPv2.2023 seawater points.

Carbonate chemistry calculations were performed using PyCO2SYS (Humphreys et al., 2022). DIC and pH on the total scale were used as input pair, together with the temperature and salinity of each observation, used to solve the carbonate-system equilibrium. Because only near-surface observations were retained, pressure was set to 0 dbar in the calculations. The main output variable used in this study was Ω_{Ar}. The carbonate equilibrium constants of Mehrbach et al. (Mehrbach et al., 1973), refit by Dickson and Millero (Dickson and Millero, 1987), were used. The bisulfate dissociation constant followed Dickson (Dickson, 1990), and total borate concentration followed Uppström (Uppström, 1974).

The baseline condition corresponded to the original GLODAPv2.2023 observations, using the measured DIC, pH, temperature, and salinity of each representative point without any added DIC. For the treatment scenarios, DIC additions of +500, +1000, and +2000 μmol kg^{-1} were applied to each representative near-surface point. In these point-based calculations, temperature, salinity, and pH were kept fixed, while DIC was increased by the prescribed amount. This approach was used to represent pH-equilibrated DIC enrichment, in which added carbon is evaluated after equilibration to the original seawater pH. Ω_{Ar} and the corresponding carbonate-system variables, including TA, were then recalculated for each representative point under the baseline condition and under the three treatment scenarios. Because DIC and pH were prescribed as the carbonate-system input pair, the recalculated TA represents the alkalinity required to maintain the imposed pH and is higher than the baseline TA in the treatment scenarios. For visualization on the salinity-temperature-pH maps, each representative point was assigned to the nearest fixed-pH panel among the selected pH values. This assignment was used only for plotting; the Ω_{Ar} values of the representative points were calculated using their measured pH values.

In addition to the point-based calculations, gridded background calculations were performed to generate the salinity-temperature-pH screening maps. For these maps, Ω_{Ar} was calculated across salinity-temperature space at fixed pH values and prescribed reference DIC values. The baseline background fields were calculated using a reference DIC value of 2000 μmol kg^{-1}, while the treatment background fields were calculated using reference DIC values of 2500, 3000, and 4000 μmol kg^{-1} for the +500, +1000, and +2000 μmol kg^{-1} scenarios, respectively. These background fields were used only for visualization and comparison with the representative GLODAP points.

For regional comparison, Ω_{Ar} values were summarized for each sea or region using the median and range of the 15 representative points. In addition, the number of representative points exceeding $\Omega_{Ar} = 7$ was counted for the +2000 μmol kg^{-1} DIC scenario. Geographic visualizations were prepared using five Ω_{Ar} ranges: $\Omega_{Ar} < 3$, $3 \le \Omega_{Ar} < 5$, $5 \le \Omega_{Ar} < 7$, $7 \le \Omega_{Ar} < 10$, and $\Omega_{Ar} \ge 10$. The same $\Omega_{Ar} = 7$ was used to show the movement of lower- and higher-supersaturation zones across the maps.

3. Results and discussion

Table 1 summarizes the baseline and treatment Ω_{Ar} values for the selected seas and ocean regions. Each value represents the median and range of the 15 representative near-surface points selected for that region. At baseline, all median Ω_{Ar} values remain below 5. However, the difference among regions is evident. The lowest median value is in the Beaufort Sea (1.15), while the highest occurs in the Red Sea (4.49). Other relatively high baseline values are observed in the Arabian Sea, Coral Sea, Sargasso Sea, South Atlantic, Philippine Sea, Mozambique Channel, and South Indian Ocean, where median Ω_{Ar} generally range from 3.6 to 4.5. In contrast, the Baffin Bay, Barents Sea, Sea of Okhotsk, Greenland Sea, and North Sea show lower baseline medians. This indicates that the carbonate system is not uniform among the selected regions. Therefore, the initial conditions in our simulations are uneven before any simulated treatment is applied. Even if we consider cold-type seawater, relevant differences are visible. For example, the Arctic regions generally have low median Ω_{Ar}, but some of them show wide ranges, especially the Greenland, Barents, and Beaufort seas.

Table 1. Regional summary of aragonite saturation state (Ω_{Ar}) under baseline conditions and pH-equilibrated DIC-addition scenarios. Values are reported as the median with the range in parentheses for 15 representative near-surface GLODAPv2.2023 points per sea or region. Treatment scenarios correspond to DIC additions of +500, +1000, and +2000 μmol kg^{-1} while temperature, salinity, and pH are kept fixed in the carbonate-system calculations. The final column reports the number of representative points exceeding $\Omega_{Ar} = 7$ under the +2000 μmol kg^{-1} DIC scenario.

Ocean	Sea/region	N points	Baseline Ω_{Ar}, median (range)	+500 Ω_{Ar}, median
Indian	Red Sea	15	4.49 (3.56–4.69)	5.58 (4.43–5.88)
Indian	Arabian Sea	15	3.93 (2.29–4.22)	4.92 (2.82–5.27)
Indian	Bay of Bengal	15	3.59 (3.38–4.05)	4.55 (4.30–5.14)
Indian	Mozambique Channel	15	3.66 (3.47–3.78)	4.58 (4.34–4.74)
Indian	South Indian Ocean	15	3.63 (3.26–4.32)	4.54 (4.05–5.41)
Pacific	Coral Sea	15	3.78 (3.55–4.00)	4.75 (4.45–5.06)
Pacific	Philippine Sea	15	3.66 (3.42–3.88)	4.59 (4.28–4.89)
Pacific	Equatorial Pacific	15	3.52 (2.51–4.04)	4.41 (3.11–5.09)
Pacific	Tasman Sea	15	3.41 (3.04–3.63)	4.25 (3.79–4.53)
Pacific	Sea of Okhotsk	15	2.05 (1.42–3.35)	2.56 (1.76–4.22)
Atlantic	Sargasso Sea	15	3.92 (3.39–4.30)	4.88 (4.19–5.36)
Atlantic	Mediterranean Sea	15	3.20 (2.71–3.70)	3.93 (3.35–4.51)
Atlantic	Bay of Biscay	15	3.05 (2.38–3.34)	3.80 (2.95–4.17)
Atlantic	South Atlantic	15	3.72 (2.65–4.80)	4.60 (3.30–6.02)
Atlantic	North Sea	15	2.36 (1.41–3.80)	2.93 (1.73–4.78)
Arctic	Greenland Sea	15	2.15 (1.21–6.07)	2.68 (1.50–7.61)
Arctic	Barents Sea	15	1.90 (1.41–5.49)	2.37 (1.73–6.93)
Arctic	Beaufort Sea	15	1.15 (0.09–5.04)	1.47 (0.13–6.75)
Arctic	Baffin Bay	15	1.64 (0.43–2.89)	2.06 (0.53–3.63)

Fig. 2 compares baseline Ω_{Ar} with pH, salinity, and temperature. The strongest correlation occurs with pH (Fig. 2a). This is expected because higher pH shifts DIC speciation toward a larger carbonate ion fraction, which directly increases Ω_{Ar} (Eq. (1)). Salinity also affects the pattern, as higher salinity water generally contains more calcium and often has higher alkalinity. Moreover, temperature has an additional effect through aragonite solubility. For comparable carbonate chemistry, warmer water has a lower aragonite solubility product, and this tends to increase Ω_{Ar}. However, Fig. 2b shows that temperature alone cannot explain the full pattern. Some polar sites with temperatures below 0 °C still reach relatively high Ω_{Ar} when pH was high. This means that other carbonate chemistry parameters such as pH are also important, especially in cold water regions. Overall, warm, saline, and high pH seas show higher Ω_{Ar}, whereas colder, lower pH, or lower salinity seas generally show lower Ω_{Ar}.

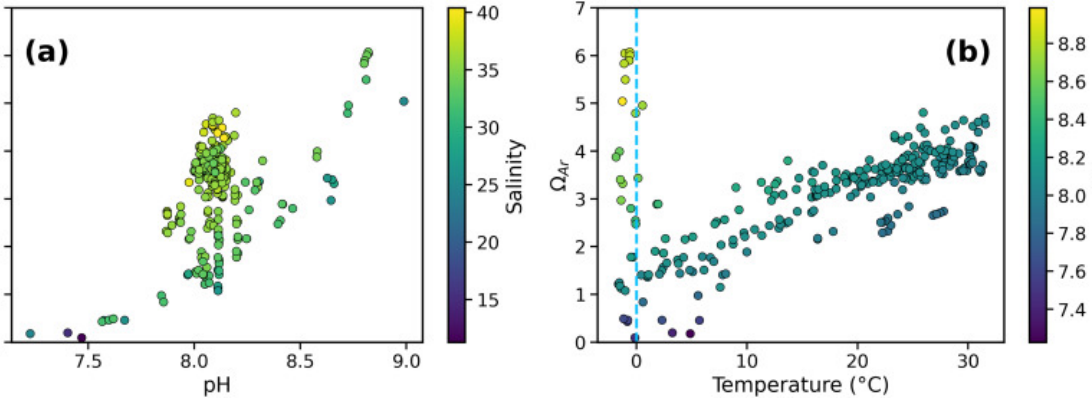

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Fig. 2. Baseline relationships between Ω_{Ar} and selected seawater properties for the representative near-surface observations. (a) Ω_{Ar} as a function of pH, with point color indicating salinity and (b) Ω_{Ar} as a function of temperature, with point color indicating pH.

Fig. 3 shows the baseline Ω_{Ar} 2D map as a function of salinity and temperature, each at a fixed pH value. The overlaid points show the spread of the selected representative data points from all seas and regions within this salinity-temperature-pH space. In each map, Ω_{Ar} increases toward higher temperature and higher salinity, while the lower temperature and lower salinity zone of the map remains farther away from the $\Omega_{Ar} = 7$ reference contour line. The shift from the low pH maps to the high pH maps is also clear: at higher pH, the same salinity-temperature conditions give higher Ω_{Ar}. This agrees with Fig. 2, where pH features the strongest correlation with baseline Ω_{Ar}.

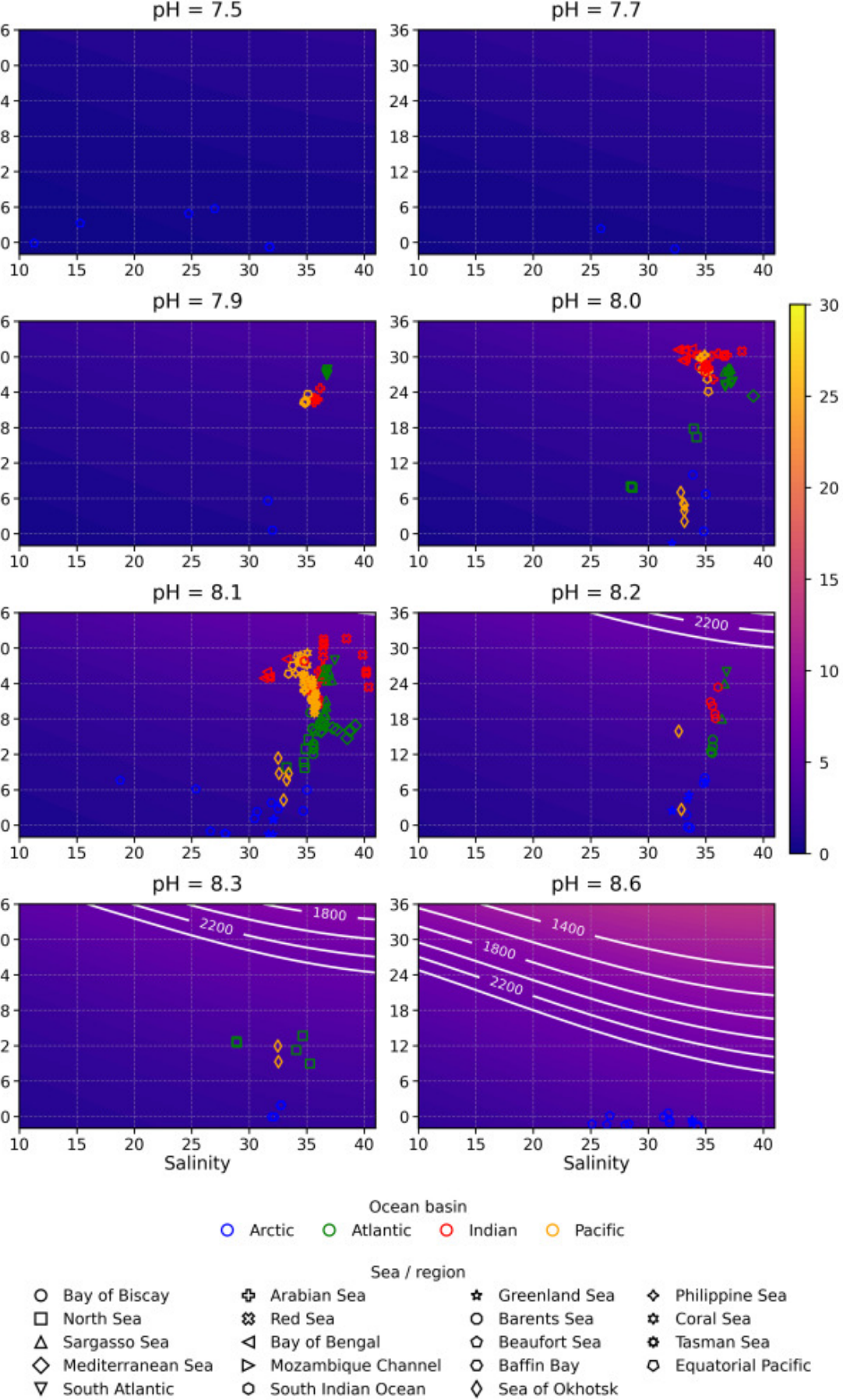

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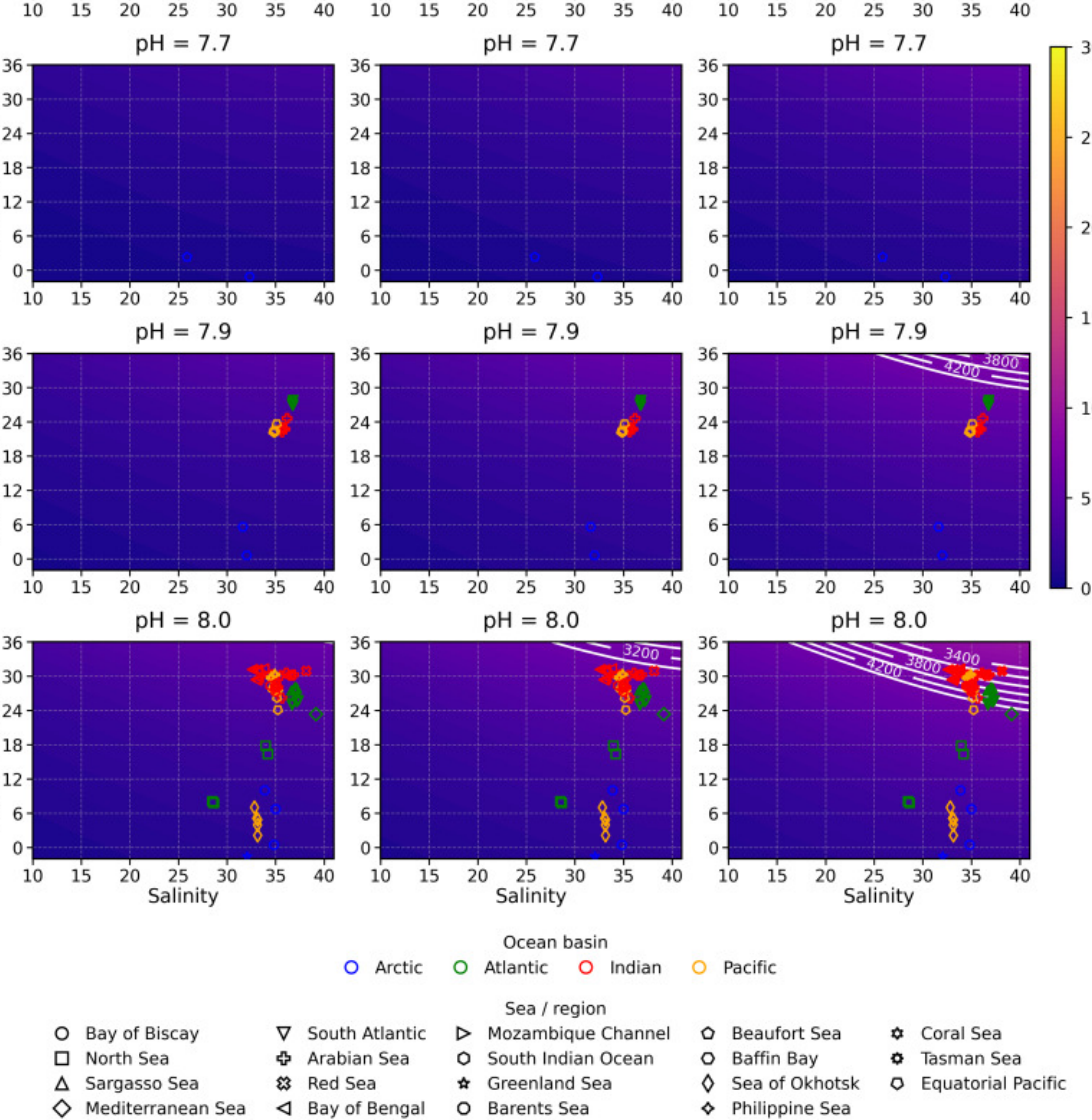
Fig. 3. Baseline carbonate-system screening of representative near-surface seawater conditions. PyCO2SYS calculations map aragonite saturation state (Ω_{Ar}) across salinity-temperature space at fixed pH values from 7.5 to 8.6. The colored background shows Ω_{Ar} calculated at a reference DIC of 2000 μmol kg^{-1}, using the same color scale in all panels. A single color bar applies to all panels. White contour lines show $\Omega_{Ar} = 7$ for DIC from 1400 to 2400 μmol kg^{-1} in steps of 200; numbers give the DIC of every second line. Representative GLODAPv2.2023 observations from the upper 10 m are plotted according to their measured salinity and temperature and assigned to the nearest fixed-pH panel for visualization. Point colors indicate ocean basins, while marker shapes identify the selected seas or regions.

The white contour lines show where the $\Omega_{Ar} = 7$ condition would be reached for different reference DIC values. For a given pH, temperature, and salinity, higher DIC moves the system closer to the $\Omega_{Ar} = 7$ reference condition, while lower DIC leaves a wider low-risk zone before this reference level is reached. Therefore, Fig. 3 shows that the baseline stability margin is not the same for all regions. The sites associated with higher temperature, higher salinity, and higher pH are closer to the high supersaturation state, whereas colder or lower salinity regions are generally farther from it. Notice that most representative points fall in the pH 8.0 to 8.2 maps, which is consistent with typical near-surface seawater conditions.

We considered three treatment scenarios, which could be representative of conditions ideally created following OAE implementation through a pH-equilibrated approach. We simulated the treatments in this study by adding a specific DIC amount to the initial (natural) DIC of each site, assuming that seawater pH remains unchanged. The addition of DIC at fixed pH increases Ω_{Ar} in all seas (Table 1). This is expected because, at a fixed pH, a larger DIC pool also gives a larger absolute carbonate ion concentration, while salinity controls calcium concentration and temperature affects the aragonite solubility product. Therefore, the increase in DIC shifts all points toward higher Ω_{Ar}, but the final Ω_{Ar} values are not the same for all regions (Fig. 4). At +500 μmol kg^{-1} added DIC, all regional median values remain below $\Omega_{Ar} = 7$. The Red Sea has the highest median value at this addition level (5.58), but it is well below the reference level. At +1000 μmol kg^{-1} added DIC, the increase becomes clearer, especially in the high-temperature, high-salinity, and high-pH regions. The Red Sea approaches the reference level, with a median Ω_{Ar} of 6.68 and a maximum value of 7.07. Some individual points in the South Atlantic and in the Greenland, Barents, and Beaufort seas also exceed $\Omega_{Ar} = 7$, although their regional medians remain below this level. This shows that a region with low median Ω_{Ar} can still have local carbonate characteristics that become sensitive under DIC addition. Fig. 4 shows this effect visually, as more points move closer to, or across, the $\Omega_{Ar} = 7$ contour on going from the +500 to the +1000 μmol kg^{-1} scenario.



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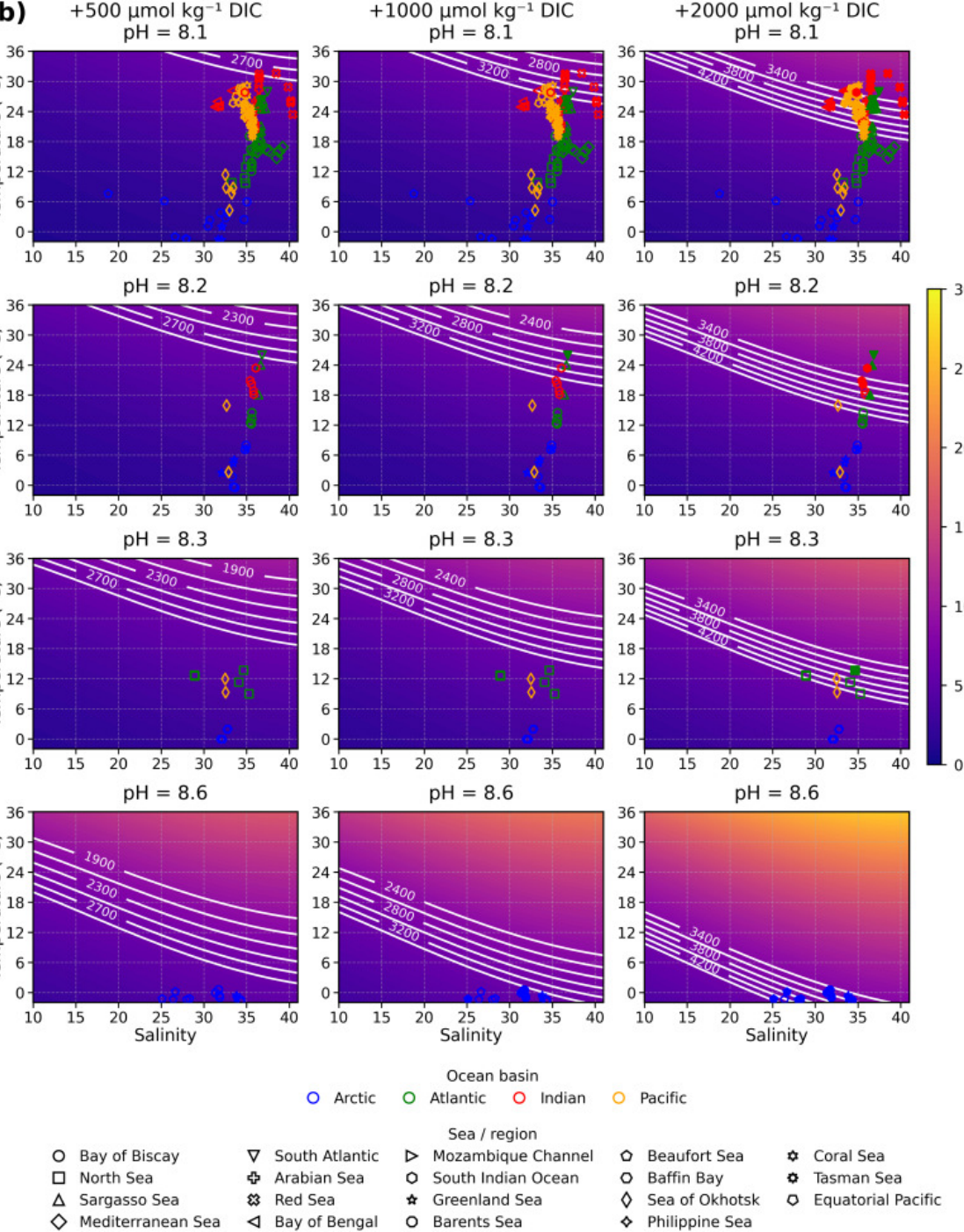

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Fig. 4. (a) pH 7.5–8.0; (b) pH 8.1–8.6. Treatment-induced shifts in aragonite supersaturation across representative seawater conditions. PyCO2SYS calculations show aragonite saturation state (Ω_{Ar}) across salinity-temperature space after DIC additions of +500, +1000, and +2000 μmol kg^{-1}. Columns correspond to the added-DIC scenarios, and rows correspond to fixed-pH panels used for visualization. The colored background shows Ω_{Ar} calculated at the scenario-specific reference DIC values of 2500, 3000, and 4000 μmol kg^{-1}, respectively. A single color bar applies to all panels. White contour lines show $\Omega_{Ar} = 7$ for DIC from 1900 to 2900 (+500), 2400 to 3400 (+1000), and 3400 to 4400 μmol kg^{-1} (+2000) in steps of 200; numbers give the DIC of every second line. Representative GLODAPv2.2023 observations are plotted according to their measured salinity and temperature and assigned to the nearest fixed-pH panel. Point colors indicate ocean basins, marker shapes identify the selected seas or regions, and filled markers denote representative points exceeding $\Omega_{Ar} = 7$ after treatment.

At +2000 μmol kg^{-1} added DIC, the shift toward higher supersaturation states becomes much stronger (Table 1, Fig. 4, and Fig. 5). Fig. 5 shows the same treatment results on a geographic map and separates the representative points into Ω_{Ar} categories to make it easier to identify sites that move into the high supersaturation state after treatment. In this scenario, several regions move clearly above $\Omega_{Ar} = 7$. All representative points exceed this threshold in the Red Sea, Bay of Bengal, and Coral Sea, and this happens also for most points in the Mozambique Channel, Philippine Sea, and Sargasso Sea. These regions therefore exhibit the strongest shift toward high-supersaturation conditions and the smallest stability margins under the imposed +2000 μmol kg^{-1} scenario. Several other regions show a mixed response. The Arabian Sea, South Indian Ocean, and South Atlantic each have 9 out of 15 points above $\Omega_{Ar} = 7$, while the Equatorial Pacific has 8 out of 15. The Greenland Sea and Barents Sea each have 5 out of 15 points above this level, the Beaufort Sea has 4 out of 15, the Tasman Sea has 3 out of 15, and the North Sea has 1 out of 15. In contrast, the Mediterranean Sea, Bay of Biscay, Baffin Bay, and Sea of Okhotsk remain below $\Omega_{Ar} = 7$ for all selected points even under the +2000 μmol kg^{-1} scenario. The Mediterranean Sea also remains below this level, although its maximum value reaches 6.94, very close to the reference level.

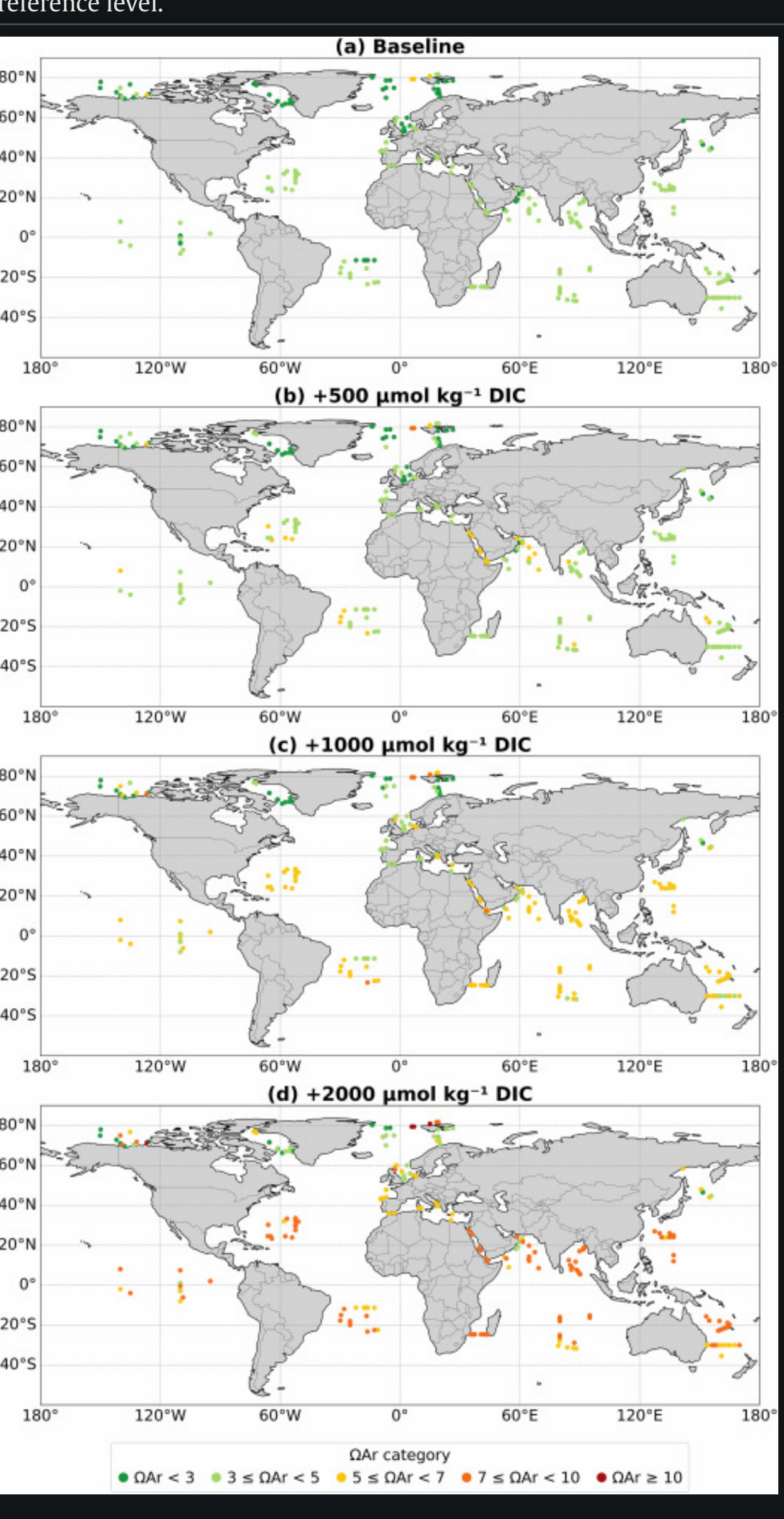

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Fig. 5. Geographic distribution of aragonite supersaturation categories under baseline and treatment conditions. Maps show Ω_{Ar} categories for the representative GLODAPv2.2023 seawater points under (a) baseline conditions, (b) +500 μmol kg^{-1} DIC, (c) +1000 μmol kg^{-1} DIC, and (d) +2000 μmol kg^{-1} DIC. Each point represents one selected near-surface observation from the sea or region shown in Fig. 1. Baseline Ω_{Ar} was calculated from the measured salinity, temperature, DIC, and pH of each representative point. Treatment scenarios were calculated by adding DIC while keeping salinity, temperature, and pH fixed. The same Ω_{Ar} categories and color scale were applied to all panels: $\Omega_{Ar} < 3$, $3 \le \Omega_{Ar} < 5$, $5 \le \Omega_{Ar} < 7$, $7 \le \Omega_{Ar} < 10$, and $\Omega_{Ar} \ge 10$.

Overall, the +2000 μmol kg^{-1} scenario shows that high DIC addition at fixed pH can strongly reduce the stability margin in some regions. In these high-supersaturation conditions, secondary $CaCO_3$ precipitation becomes more likely. This interpretation is supported by previous mesocosm experiments on pH-equilibrated OAE (Jamali Alamooti et al., 2026). Jamali Alamooti et al. (Jamali Alamooti et al., 2026) showed that excessive increases in Ω_{Ar} following OAE treatment can strongly reduce stability shortly after discharge. In highly supersaturated treatments, secondary $CaCO_3$ precipitation and CO_2 degassing caused rapid losses of added DIC and alkalinity. In some cases, DIC and alkalinity even decreased below the initial seawater values before treatment because once the carbonate precipitation is triggered, it continues even for Ω_{Ar} values for which no precipitation would start. Therefore, crossing into very high Ω_{Ar} conditions is not only a chemical indicator of supersaturation, but also a warning for possible loss of OAE efficiency and broader negative impacts on the marine environment.

The +2000 μmol kg^{-1} scenario should be considered as a high-end, near-field, screening case before substantial plume dilution, rather than as a far-field basin-scale prediction. In real discharge conditions, whether secondary $CaCO_3$ precipitation occurs would depend not only on Ω_{Ar}, but also on how rapidly the treated plume mixes and dilutes in the receiving seawater. If dilution reduces Ω_{Ar} before precipitation is initiated, the actual precipitation risk may be lower than indicated by the static carbonate-system screen. Conversely, slow dilution or locally persistent high supersaturation could increase the probability of precipitation.

It should also be noted that the present maps do not represent fixed seasonal boundaries. The selected observations come from different sampling dates and seasons, and seasonal changes in temperature, salinity, pH, DIC, and air-sea CO_2 conditions can shift Ω_{Ar}. This effect may be more relevant in high latitude and coastal regions, where freshwater input, sea-ice influence, biological activity, or upwelling can strongly change carbonate chemistry. Therefore, the maps should be used as a regional screening of relative stability margins, while site-specific application will require seasonal or real-time local carbonate chemistry monitoring.

4. Conclusion

This study estimated the effect of seawater chemistry diversity on pH-equilibrated OAE, with a focus on aragonite supersaturation and the correlated carbonate precipitation risk. Representative near-surface observations from selected sea and ocean regions were obtained from GLODAPv2.2023 and used as input parameters for PyCO2SYS calculations with the purpose of predicting the change of supersaturation. Baseline Ω_{Ar} was first calculated from the natural seawater properties. Thereafter, three pH-equilibrated OAE scenarios were considered by imposing constant DIC enrichments of +500, +1000, and +2000 μmol kg^{-1}, corresponding to approximately 22, 44, and 88 g CO_2-equivalent per ton of seawater, while keeping pH, temperature, and salinity fixed (using the CO_2 molar mass of 44.01 g mol^{-1} and kg to ton conversion). In each scenario, the DIC addition was the same for all regions, so the differences in final Ω_{Ar} reflect the initial seawater chemistry.

The analysis starts from the observation that the initial carbonate system state is not uniform among different regions. The hypothesis of this work is that those inhomogeneities strongly affect OAE treatments, including the pH-equilibrated one. As a matter of fact, high-temperature, high-salinity, and high-pH regions are generally in a higher supersaturation state for aragonite, whereas colder or lower-salinity regions generally features lower Ω_{Ar}. An addition of DIC at fixed pH increases Ω_{Ar}. Because the regions started from different baseline conditions, the same DIC addition leads to different final supersaturation states and therefore different stabilities. At moderate addition levels (+1000 μmol kg^{-1}), most regional median values remain below $\Omega_{Ar} = 7$. In contrast, the +2000 μmol kg^{-1} scenario pushed several regions into higher-supersaturation conditions.

From a deployment perspective, these results indicate that pH-equilibrated OAE should not be planned using a universal DIC or alkalinity dosage. Region-specific carbonate-system assessment is fundamental before selecting suitable treatment levels, especially in sites that are already close to high aragonite supersaturation. Previous mesocosm experiments support this interpretation, showing that excessive Ω_{Ar} after OAE treatment may lead to secondary $CaCO_3$ precipitation, CO_2 degassing, and loss of added DIC and alkalinity. At the same time, the present results should be interpreted as a regional assessment rather than a time/season-specific deployment prediction, because the selected observations come from different dates and seasons. Consequently, the boundaries between lower and higher supersaturation conditions may shift with seasonal changes in temperature, salinity, or CO_2 atmospheric pressure, especially in coastal regions. Real OAE discharge would also be affected by plume mixing and dilution, as well as discharge depth, biological activity, temperature changes, and air-sea CO_2 exchange.

This study provides useful input for the selection of best sites for pH-equilibrated OAE, in terms of efficiency of the stability of the stored bicarbonates. Of course, other considerations are necessary before undertaking a site selection, such as economics, life-cycle assessment, geographical suitability, etc. Therefore, this study is a tassel of a bigger mosaic that could be completed merging the different aspects of OAE methods. Nonetheless, this approach could be used also for assessing the suitability of other OAE methods, provided that all natural parameters are available. For non-equilibrated OAE, for example, the pH increase upon treatment should be explicitly considered.

CRediT authorship contribution statement

Samira Jamali Alamooti: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Guido Raos:** Writing – review & editing, Validation, Supervision. **Piero Macchi:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The PhD project of Samira Jamali Alamooti was financially supported by the Italian Ministry of University and Research (MUR) and by Limenet Srl. The authors declare no other known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Limenet Srl had no direct role in the design of the present analysis, data processing, interpretation of the results, or writing of the manuscript.

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Data availability

The GLODAPv2.2023 data used in this study are publicly available from the GLODAP data product. The processed representative-point dataset, PyCO2SYS calculation outputs, and scripts used to generate the results and figures will be made available on request.

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